Logistic Regression under Missing Data using Tractable Circuits

Goal:

Given $x, w \in \mathbb{R}^d$ (data and model weights), we wish to compute

$$\mathbb{E}_{p(x)}[\sigma(w^\top x)]$$

efficiently, where p is PC-distribution over x.

We will then use this to compute expectations under *missing features* in x.

Step 1: PG identity for the sigmoid

Reference: arXiv

Use the Pólya–Gamma (PG) identity (Theorem 1 of Polson–Scott–Windle) to reduce $\mathbb{E}_{p(x)}[\sigma(w^\top x)]$ to **nested 1-D integrals** that you can evaluate on a PC without mixture expansion.

Theorem 1 states that for $b > 0$, $a \in \mathbb{R}$, $\kappa = a - \frac{b}{2}$, and $\omega \sim \mathrm{PG}(b, 0)$,

$$\frac{e^{a\psi}}{(1+e^{\psi})^b} = 2^{-b} e^{\kappa\psi} \int_0^\infty e^{-\frac{1}{2}\omega\psi^2}\, p_{\mathrm{PG}(b,0)}(\omega)\, d\omega.$$

For the logistic sigmoid $\sigma(\psi) = e^{\psi}/(1+e^{\psi})$ choose $a = b = 1 \Rightarrow \kappa = \frac{1}{2}$:

$$\sigma(\psi) = \frac{1}{2}\, e^{\psi/2}\, \mathbb{E}_{\omega \sim \mathrm{PG}(1,0)}\left[e^{-\frac{1}{2}\omega\psi^2}\right]. \quad \text{(exact)}$$

Set $\psi = w^\top x$. Then

$$\boxed{\mathbb{E}_{p(x)}[\sigma(w^\top x)] = \frac{1}{2}\, \mathbb{E}_{\omega \sim \mathrm{PG}(1,0)}\Big[\underbrace{\mathbb{E}_{p(x)}\big[e^{-\frac{1}{2}\omega(w^\top x)^2 + \frac{1}{2}(w^\top x)}\big]}_{=: G(\omega)}\Big].}$$

This moves the nonlinearity into an **exponentiated quadratic** of $w^\top x$.

Step 2: Evaluate $G(\omega)$ on a PC without expansion (1-D Hermite via CF)

Let $Y := w^\top x$. For **any** smooth, decomposable PC with Gaussian leaves you can evaluate the **characteristic function**

$$\phi_Y(s) = \mathbb{E}_{p(x)}[e^{isY}]$$

in $O(|\mathrm{PC}|)$ by a single bottom-up pass:

- Leaf (Gaussian over scope S): $\phi(s) = \exp\big(is\, w_S^\top \mu - \frac{1}{2} s^2\, w_S^\top \Sigma w_S\big)$.
- Sum node: $\phi = \sum_j \alpha_j \phi_j$.
- Product node (disjoint scopes): $\phi = \prod_j \phi_j$.

Now use the Gaussian–Fourier identity (complete the square, then Fourier-invert) to write, for $t \in \mathbb{R}$,

$$\mathbb{E}\big[e^{-\frac{1}{2}\omega Y^2 + tY}\big] = \frac{e^{t^2/(2\omega)}}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-z^2}\, e^{-izt\sqrt{2/\omega}}\ \phi_Y\big(\sqrt{2\omega}\, z\big)\, dz.$$

Hermite-approximate the 1-D integral with nodes/weights $\{(z_i, w_i)\}_{i=1}^N$:

$$\boxed{G(\omega)\ \approx\ \frac{e^{t^2/(2\omega)}}{\sqrt{\pi}} \sum_{i=1}^{N} w_i\, e^{-iz_i t\sqrt{2/\omega}}\ \phi_Y\big(\sqrt{2\omega}\, z_i\big), \qquad t = \tfrac{1}{2}.}$$

This uses only $\phi_Y(\cdot)$, hence only **PC-local recurrences**—no GMM expansion.